CrowdWisdomTrading Quantitative Assessment

Asset Price Prediction Agent via X (Twitter) Sentiment & Price | Walk-Forward Validation

NOTICE: Evaluated on synthetic demonstration dataset for pipeline verification. Metrics confirm zero lookahead & execution validity.

1. System Architecture & Model Setup

Target Asset: Bitcoin (BTCUSDT) | **Bar Size:** 1h | **Total Bars:** 4,249 (177.0 days)
Forecast Horizons: n+0 through n+5 hourly bars ahead.
Features: Strictly restricted to X sentiment (mean, engagement-weighted, ratios, volume z-score) and price (lagged returns, rolling volatility, range, intraday & day-of-week cycles).
Model Engine: Ensemble (L2-Regularised Ridge + HistGradientBoosting Regressor).
Validation Strategy: Expanding-window walk-forward (6 folds) with strict 6-bar purge window between training and test sets to eliminate target overlap leakage.

2. Out-of-Sample Risk-Adjusted Performance

Performance Metric	Strategy (Sentiment + Price)	Benchmark (Buy & Hold)
Annualised Sharpe Ratio	4.03	-0.84
Annualised Sortino Ratio	5.89	-1.18
Maximum Drawdown (MaxDD)	-15.7%	-36.8%
Total Out-of-Sample Return	65.8%	-9.9%
Trade Win Rate	51.9%	49.2%
Turnover Per Bar	0.59	0.00

3. Multi-Horizon Forecast Accuracy (n+0 to n+5)

Horizon	MAE ($)	Naive MAE ($)	Skill vs Naive	Dir. Accuracy	Spearman IC
n+0	$88.5	$89.3	+0.94%	54.8%	+0.133
n+1	$128.6	$130.2	+1.25%	56.4%	+0.164
n+2	$158.8	$161.5	+1.69%	57.1%	+0.199
n+3	$183.2	$188.8	+2.96%	58.6%	+0.241
n+4	$203.7	$211.9	+3.89%	59.5%	+0.264
n+5	$225.0	$235.9	+4.62%	60.0%	+0.282

4. Walk-Forward Fold Consistency

Fold 0: Sharpe -6.17 | **Fold 1:** Sharpe 2.95 | **Fold 2:** Sharpe 0.43 | **Fold 3:** Sharpe 8.94 | **Fold 4:** Sharpe 2.90 | **Fold 5:** Sharpe 15.84

5. Rigorous Leakage & Integrity Controls

• **Zero Lookahead:** Feature bar aggregation floors timestamp to bar open; models only ingest data available at bar close.
• **Purged Walk-Forward:** Between training window and out-of-sample test window, an embargo gap equal to maximum horizon (6 bars) is dropped.
• **Transaction Cost Model:** All performance metrics deduct 5 bps per unit of turnover to simulate realistic slippage and exchange fees.
• **Independent Targets:** Models forecast log-returns from current close to each horizon independently.